

Experience

- TaskVeda** Remote
AI Deep Learning Intern Jul 2025 – Aug 2025
 - Fine-tuned **InLegalBERT** (a BERT variant pretrained on Indian legal corpora) for domain-specific question answering using Python, PyTorch, and the Hugging Face Transformers library
 - Designed and implemented evaluation pipelines using ROUGE, BLEU, and exact-match metrics to systematically benchmark model performance across 5K+ legal queries
- IGDTUW 3D Map Project** Delhi
Blender Team Lead Nov 2024 – Jan 2025
 - Led and coordinated an 8-member team to reconstruct the IGDTUW campus as a photorealistic 3D model using Blender, adhering to architectural blueprints
 - Architected the asset pipeline, assigning modular building components, managing version control via shared drives, and conducting weekly design reviews
 - Delivered a fully navigable 3D campus map used for virtual tours and university promotional content

Projects

- SIMx1 | C++17, CMake, GTest** Ongoing
 - Developed a modular, object-oriented gate-level logic simulator from scratch in C++17, implementing primitive gates (AND, OR, NOT, NAND, NOR, XOR) as polymorphic classes with unified interfaces
 - Implemented a 4-bit ripple-carry adder by composing primitive gates into hierarchical modules, and verified correctness across all 256 input combinations using a GoogleTest-based self-checking testbench
 - Currently extending the simulator with D-flip-flop and JK-flip-flop support to enable sequential logic simulation, along with waveform dumping in VCD format for visual debugging
- RAG Janta AI | Python, InLegalBERT, FAISS, LangChain** August 2025 - October 2025
 - Built a Retrieval-Augmented Generation (RAG) system for legal document retrieval, combining a dense vector index (FAISS) with InLegalBERT for semantically-aware search across 10K+ Indian court judgments
 - Fine-tuned InLegalBERT on a custom dataset of legal question-answer pairs using contrastive learning to improve retrieval relevance (achieved 12% improvement in recall@10)
 - Integrated the retriever with a generative LM via LangChain, implementing a hybrid search strategy (dense + sparse) to balance precision and recall for legal information access

Technical Skills

- Languages:** C++, Python, JavaScript (ES6+), SQL (PostgreSQL, MySQL), C#, HTML5, CSS3
- Frameworks & Libraries:** React.js, Node.js, Express.js, PyTorch, Hugging Face Transformers, LangChain, FAISS
- Developer Tools:** Git, GitHub, Linux/Bash, CMake, GoogleTest, Docker, Unity, Blender

Achievements & Awards

- Top 5 Finalist (out of 1500+ participants)** – HackArena, IIIT Delhi for building an innovative a roblox game with help of lua scripts in a 36-hour hackathon
- Top 12 Finalist (out of 600+ teams)** – Design With India, DTU for creating a sustainable product design prototype
- Top 40 (out of 3000+ participants)** – Algoverse Hackathon, competing against national-level talent in algorithmic problem solving
- 1st Position** – Uxcelerate UI/UX Hackathon for designing a user-centric mobile application interface
- 3rd Position** – ML vs LLM Hackathon for developing a hybrid machine learning pipeline for real-world data classification
- Semi-Finalist** – Great AppSec Hackathon 2024 for building a security-focused web application with vulnerability detection features

Leadership & Extracurricular

- Media Lead, Taarangana (IGDTUW's Cultural Fest)** – Produced and executed social media campaigns that generated 40K–80K+ organic views across Instagram and LinkedIn; managed a team of 12 content creators and designers
- Game Development Mentor** – Conducted weekly workshops and mentored 60+ students in building 3 complete game projects using Unity and C# over a 4-month period; Helping build 1 action 3D Game alongwith two 2D games
- Head of International Press, SRDAVMUN** – Led a team of 50 student journalists covering international Model United Nations conferences; mentored delegates whose teams collectively secured 30+ awards at national-level MUN events

Education

- Indira Gandhi Delhi Technical University for Women** Delhi, India
B.Tech in Electronics & Communication Engineering (AI Specialization) Aug 2024 – Present